UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION- SEMESTER I (2018/2019)
CS 101 – Introduction to Computer Science (3 Credits)

Time Allowed: **TWO** Hours Answer **ALL** Questions

1.

a. Briefly describe the contributions of the following individuals to the computing domain.
[06 marks]

i. Charles Babbage
ii. Ada Loveless
iii. John von Neuman

b. Explain the terms *Diligence* and *Versatility* with reference to characteristics of a computer.
[04 marks]

c. Illustrate the block diagram of the Von Neuman architecture and explain the functionality of each of the units.
[08 marks]

d. Explain how the statement *c = a + b;* in a C programme, where a,b and c are integers would be handled in the Fetch-Execute cycle of a computer.
[07 marks]

2.

a. Briefly explain three (03) key parameters of a CPU.
[06 marks]

b. What category of memory do the following memory components fall under?
[03 marks]

i. BIOS chip
ii. Main memory
iii. Internal Hard Disk drives

c. What are the main factors that one would consider when purchasing a computer monitor?
[03 marks]

d. Computer software can be categorized according to their functionality.

i. Using a diagram illustrate how these different categories of software are layered/stacked between the Human-User and the Computer-Hardware.
[03 marks]

ii. Provide at least two (02) examples for each category of software illustrated in the above diagram.
[03 marks]

e. What is the Octal and Hexadecimal representation of 596_{10}?
[04 marks]

f. What is the decimal representation the following?
[03 marks]

i. 10011011_2 sign-magnitude representation
ii. 01111010_2 1s compliment representation

iii. 11101110_2 signed 2s compliment representation

3.

a. Perform the following binary arithmetic operations (clearly indicate how the operations were performed).

[06 marks]

i. 1100000 – 10000010
ii. 1100000 / 100
iii. 1100000 x 101

b. Perform a BCD addition of the following numbers 1001000010_{BCD} and 101101001_{BCD}?

[02 marks]

c. Represent 72.6875_{10} in IEEE754 floating point representation.

[02 marks]

d. Perform the following tasks.

i. Define what is an algorithm.

[02 marks]

ii. Represent 156_{10} in binary

[02 marks]

iii. Write an algorithm in pseudocode that would take any positive integer value (decimal) and will display a binary output.

[11 marks]

4. You are required to automate the control of the water tank in your home. There are two water contact sensors that turn to a TRUE state when in contact with water and a FALSE state when not. Sensor 1 (S1) to detect water at the maximum level of the tank and sensor 2 (S2) detects water level at the minimum level. The electronically controlled tap (T) must turn off (FALSE) when the water level reaches S1 and the tap turn on (TRUE) when the water level drops to S2. There is a State Register (SR) which changes state if and only if S1, S2 and SR are all the same state (i.e. S1 TRUE, S2 TRUE and SR TRUE then SR changes to FALSE or S1 FALSE, S2 FALSE and SR FALSE then SR changes to TRUE).

a. Consider the output of this system as the state of T. Note down the possible states of this system in a truth table.

[04 marks]

b. Write down the SOP form of the above truth table.

[04 marks]

c. Using Boolean algebra simplify the SOP into its minimal form (note what laws and postulates you use).

[04 marks]

d. Draw the Karnaugh map for the above truth table.

[04 marks]

e. Using the Karnaugh map deduce the minimal POS form.

[04 marks]

f. Design a circuit for the above using only NOT and NOR gates.

[05 marks]